

LS INVENTORY

Panduan Instalasi, Konfigurasi & Operasional

Arsitektur Multi-Cabinet - Central Server - Device Pairing - License - Maintenance

Versi	1.0 - 29 Agustus 2026
Server Utama	Raspberry Pi 1 + HMI Lab Cabinet
Device Utama	LS_Cab_LabHMI
Central API	Port 3100

Dokumen ini menjelaskan instalasi Raspberry Pi utama dan cabinet tambahan, konfigurasi RC522/kamera/TFT, Tailscale, Web Admin, pairing device, lisensi Trial/Lifetime, sinkronisasi offline, backup, dan troubleshooting.

1. Ringkasan Sistem

LS Inventory adalah sistem inventory multi-cabinet berbasis Raspberry Pi. Raspberry Pi 1 saat ini berfungsi ganda sebagai server utama (Central) dan Cabinet HMI Lab. Raspberry Pi berikutnya hanya menjadi cabinet client dan berkomunikasi ke Central melalui API privat/Tailscale.

Komponen	Peran
Raspberry Pi 1	Central Server + HMI Lab Cabinet
Central :3100	Web Admin, Central API, central.db
Cabinet :3000	Kiosk, NFC, QR, Borrow/Return/Consume
device.db	Database lokal pada setiap cabinet
Tailscale	Jaringan privat antar-device

Arsitektur

WEB ADMIN	RASPI UTAMA	LICENSE SERVER
Browser Admin	Central :3100 central.db HMI Cabinet :3000	logisourcedigital.web.id
CABINET 2 / 3 / ...	TAILSCALE	OPERASI LOKAL

Prinsip penting: central.db hanya berada di server utama. Setiap cabinet memiliki device.db sendiri agar tetap dapat bekerja saat jaringan/Central sementara tidak tersedia.

2. Persyaratan Hardware

- Raspberry Pi dengan 40-pin GPIO, CSI camera, dan jaringan stabil.
- RC522 NFC reader 13.56 MHz pada SPI1.
- Camera CSI (OV5647 atau yang kompatibel dengan rpical).
- TFT 3.5 inch ILI9486/tft35a pada SPI0 (sesuai cabinet saat ini).
- microSD minimal 32 GB high-endurance/Class 10; 64 GB lebih disarankan.
- Power supply berkualitas sesuai model Raspberry Pi.
- Tailscale untuk koneksi privat antar-cabinet dan Central.

Rekomendasi Model Raspberry Pi

Model	Rating	Reason
Raspberry Pi 4 Model B 2GB/4GB	RECOMMENDED	Best balance: more RAM, USB 3, Gigabit Ethernet, mature CSI/GPIO stack.
Raspberry Pi 5 2GB/4GB	RECOMMENDED / FUTURE	Fastest option; requires good power/cooling. Validate existing TFT overlay/camera cable compatibility.
Raspberry Pi 3 Model B+ 1GB	MINIMUM SAFE	Compatible architecture and 40-pin GPIO; slower but acceptable for a cabinet client.
Raspberry Pi 3 Model B 1GB	WORKS	Current primary hardware proves it can run; not the preferred choice for a new purchase.

Raspberry Pi 3 Model A+ 512MB	NOT RECOMMENDED	No Ethernet, one USB port, only 512MB RAM.
Raspberry Pi Zero 2 W 512MB	NOT RECOMMENDED	512MB RAM, no Ethernet, one USB OTG; Chromium + camera + Node workload is tight.
Raspberry Pi 2 Model B	NOT RECOMMENDED	Older platform; revisions 1.1/1.2 reached EoL in 2025, no onboard Wi-Fi, limited headroom.

Kesimpulan pembelian: pilih Raspberry Pi 4 Model B 2GB atau 4GB untuk cabinet #2/#3 karena paling seimbang. Raspberry Pi 5 cocok jika ingin future-proof, tetapi perlu PSU/cooling yang lebih baik dan verifikasi kompatibilitas display/camera connector. Model di bawah Raspberry Pi 3B tidak disarankan untuk build produksi LS Inventory.

3. Wiring Hardware Cabinet

RC522 - SPI1

RC522	Raspberry Pi
3.3V	Pin 1 / 3.3V
GND	Pin 6 / GND
SCK	GPIO21 / Pin 40 / SPI1 SCLK
MOSI	GPIO20 / Pin 38 / SPI1 MOSI
MISO	GPIO19 / Pin 35 / SPI1 MISO
SDA / SS	GPIO16 / Pin 36 / CS
RST	3.3V
IRQ	Not used

```
dtparam=spi=on
dtoverlay=tft35a:rotate=0
dtoverlay=spil-1cs,cs0_pin=16
```

Jangan pindahkan RC522 ke SPI0 karena SPI0 digunakan TFT/touch controller.

4. Instalasi Raspberry Pi Utama (Pi 1)

Pi 1 menjalankan dua process PM2: LS_Inventory (Cabinet :3000) dan LS_Inventory_Central (Central :3100).

```
sudo apt update
sudo apt install -y git curl python3 python3-spidev python3-opencv chromium xinit

sudo mkdir -p /opt/LS_Inventory
sudo chown -R $USER:$USER /opt/LS_Inventory
git clone https://github.com/fbrhrdynt/LS_Inventory.git /opt/LS_Inventory
cd /opt/LS_Inventory
npm install
```

.env

```
APP_NAME="LS Inventory"
PORT=3000
DEVICE_ID=LS_Cab_LabHMI
DEVICE_NAME="Cabinet Lab HMI"
DEVICE_API_KEY=<DEVICE_KEY>
CENTRAL_API_URL=http://127.0.0.1:3100
CENTRAL_PORT=3100
```

PM2

```
cd /opt/LS_Inventory
pm2 start central/server.js --name LS_Inventory_Central --cwd /opt/LS_Inventory
pm2 start app.js --name LS_Inventory --cwd /opt/LS_Inventory
pm2 save
curl http://127.0.0.1:3100/health
```

5. Instalasi Cabinet Baru (Pi 2, Pi 3, ...)

Cabinet baru meng-clone repository yang sama, tetapi TIDAK menjalankan central/server.js dan TIDAK menyalin central.db/device.db/.env dari Pi lain.

```
sudo mkdir -p /opt/LS_Inventory
sudo chown -R $USER:$USER /opt/LS_Inventory
git clone https://github.com/fbrhrdynt/LS_Inventory.git /opt/LS_Inventory
cd /opt/LS_Inventory
npm install
```

Tailscale

```
tailscale status
curl http://<PRIMARY_PI_TAILSCALE_IP>:3100/health
```

Pairing

1. Web Admin > Devices & Maintenance > Add Device.
2. Buat DEVICE_ID unik, misalnya LS_Cab_Workshop01.
3. Ambil pairing code 6 digit (single-use, expired).
4. Pada Pi baru jalankan node scripts/pair_device.js.
5. Masukkan Central URL/Tailscale dan pairing code.
6. Pi menyimpan DEVICE_API_KEY sendiri dan menjalankan initial sync.

```
cd /opt/LS_Inventory
node scripts/pair_device.js
node services/google_sync.js
pm2 start app.js --name LS_Inventory --cwd /opt/LS_Inventory
pm2 save
```

Catatan: nama services/google_sync.js dipertahankan untuk kompatibilitas, tetapi fungsi aktualnya sekarang adalah Central Sync, bukan Google Sheets.

6. Web Admin dan Role

Menu	Administrator	Admin Staff
Dashboard	Yes	Yes
Inventory	Yes	Yes
Users	Yes	Yes
Transactions	Yes	Yes

Devices & Maintenance	Yes	No
Admin Users	Yes	No
Settings	Yes	No

Administrator dapat membuat akun Web Admin dengan role Administrator atau Admin Staff. Admin Staff dibatasi di UI dan API.

Inventory delete

Delete item menggunakan soft-delete. Riwayat transaksi tetap disimpan. Item BORROWED harus dikembalikan sebelum dapat dihapus.

7. License - Trial dan Lifetime

Setiap Raspberry Pi cabinet memiliki fingerprint hardware dan license sendiri. Raspberry Pi baru otomatis masuk TRIAL jika belum memiliki license Lifetime. Current build menggunakan trial 14 hari secara default.

Plan	Behavior
TRIAL	Full operational access while active; expires according to configured trial days.
LIFETIME	Persistent full access after successful activation/verification.

License dikelola dari Devices & Maintenance > License Details dan hanya Administrator yang dapat memasukkan license. Untuk Pi 2/3 yang remote, activation command dikirim melalui Central dan diproses oleh device target.

Endpoint license: `logisourcedigital.web.id/api/public/license/verify` dan `/activate`. Jangan commit license key ke GitHub.

8. Operasi Cabinet

- NFC Login melalui RC522.
- QR scan menggunakan UUID item.
- Manual CODE menggunakan ITEM_00000 sampai ITEM_99999 (5 digit).
- BORROWABLE: Borrow -> BORROWED; Return -> AVAILABLE.
- CONSUMABLE: qty mengurangi stock; LOW_STOCK/OUT_OF_STOCK mengikuti threshold.
- Pending transaction tetap tersimpan lokal jika Central offline.

9. Sinkronisasi dan Offline Mode

Master data mengalir dari Central ke device.db secara incremental. device.db kemudian mengekspor cache JSON untuk lookup cepat. Jika Central tidak tersedia, cabinet tetap menggunakan data lokal dan transaction queue.

```
cd /opt/LS_Inventory
node services/google_sync.js

# health
curl http://127.0.0.1:3000/health
curl http://<CENTRAL_HOST>:3100/health
```

10. Backup & Restore

Sebelum update, backup database. Database dan .env tidak boleh masuk GitHub.

```
mkdir -p /opt/LS_Inventory/backups
cp /opt/LS_Inventory/central/data/central.db /opt/LS_Inventory/backups/central-$(date +%Y%m%d-%H%M%S).db
cp /opt/LS_Inventory/data/device.db /opt/LS_Inventory/backups/device-$(date +%Y%m%d-%H%M%S).db
```

11. Maintenance Storage & Camera

Pada Pi utama pernah ditemukan puluhan core dump rpicas-vid (~287 MB per file) yang memenuhi disk hingga >90%. Core dump dapat dihapus setelah diidentifikasi dan core generation sebaiknya dinonaktifkan pada appliance produksi.

```
cd /opt/LS_Inventory
find . -maxdepth 1 -type f -name 'core.*' -print -delete
sync
df -h /

sudo tee /etc/security/limits.d/99-lsinventory-nocore.conf >/dev/null <<'EOF'
logisource soft core 0
logisource hard core 0
EOF
```

Recommended disk thresholds

Disk Usage	Status
< 75%	NORMAL
75-84%	WARNING
85-89%	HIGH
>= 90%	CRITICAL

12. Safe GitHub Workflow

Repository: github.com/fbrhrdynt/LS_Inventory. Selalu periksa staged files sebelum push.

```
cd /opt/LS_Inventory
git status --short
git diff --cached --name-only

# NEVER stage:
# .env
# data/device.db
# central/data/central.db
# data/license.json
# core.*

git add README.md docs assets scripts services central app.js views public package.json package-lock.json
git diff --cached --name-only
git commit -m "Update LS Inventory documentation and production assets"
git push origin main
```

.gitignore minimum

```
.env
node_modules/
data/*
!data/.gitkeep
central/data/*
!central/data/.gitkeep
cache/*.json
*.db
*.db-wal
*.db-shm
core.*
backups/
*.tar.gz
*.zip
*.log
```

13. Checklist Cabinet Baru

- ☐ Unique DEVICE_ID
- ☐ Unique device credential / pairing complete
- ☐ Tailscale connected
- ☐ Central :3100 reachable
- ☐ SPI1 /dev/spidev1.0 exists
- ☐ RC522 reports Version 0x92
- ☐ Camera works
- ☐ TFT works
- ☐ Initial Central sync complete
- ☐ NFC login works
- ☐ QR UUID works
- ☐ 5-digit CODE works
- ☐ Borrow/Return/Consume tested
- ☐ License status visible
- ☐ PM2 saved for boot

14. Referensi Hardware Resmi

- Raspberry Pi 4 Model B specifications - raspberrypi.com/products/raspberry-pi-4-model-b/specifications/
- Raspberry Pi 5 specifications - raspberrypi.com/products/raspberry-pi-5/
- Raspberry Pi 3 Model B+ - raspberrypi.com/products/raspberry-pi-3-model-b-plus/
- Raspberry Pi computer hardware documentation - raspberrypi.com/documentation/computers/raspberry-pi.html
- Raspberry Pi Zero 2 W - raspberrypi.com/products/raspberry-pi-zero-2-w/

Catatan kompatibilitas: pada Raspberry Pi 5, konektor MIPI camera/display berbeda dari Pi 3/4, sehingga kabel/adaptor dan overlay TFT yang digunakan saat ini harus diuji sebelum deployment.

15. Informasi Branding

LogiSource Digital

Digital. Reliable. Connected

Logo HD tersedia di assets/LogiSource_Digital_Logo_HD.png dan dapat ditampilkan pada README GitHub. File branding concept LS Inventory juga disertakan sebagai draft visual, bukan pengganti identitas utama tanpa persetujuan brand.